

Worksheet: Profit and the Price of Cheerios

ECON 201

Work with a neighbor. Write your answers in the spaces.

Three definitions you will use throughout this module. *Total revenue* is the number of units sold times the price per unit: $R = P \times Q$. *Total cost*, written $C(Q)$, is the cost of producing Q units. *Profit* is total revenue minus total cost:

$$\text{Profit} = \underbrace{P \times Q}_{\text{revenue, } R} - \underbrace{C(Q)}_{\text{total cost}}$$

1. Do higher prices always mean higher profits?

A firm's profit is $P \times Q - C(Q)$. To maximize profit, why not just charge a really high price? Jot down your thoughts, then compare answers with a neighbor.

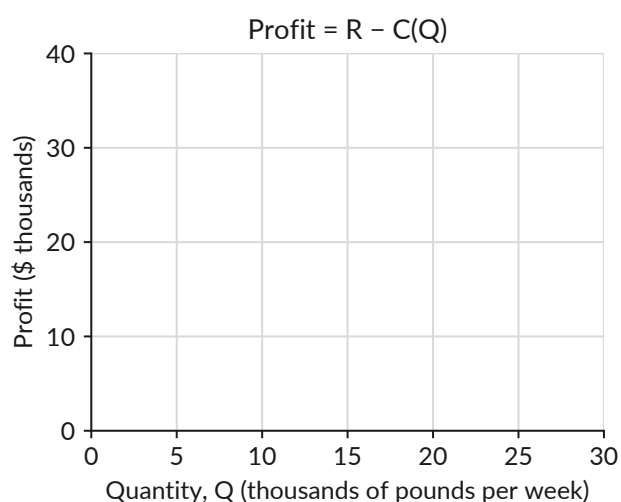
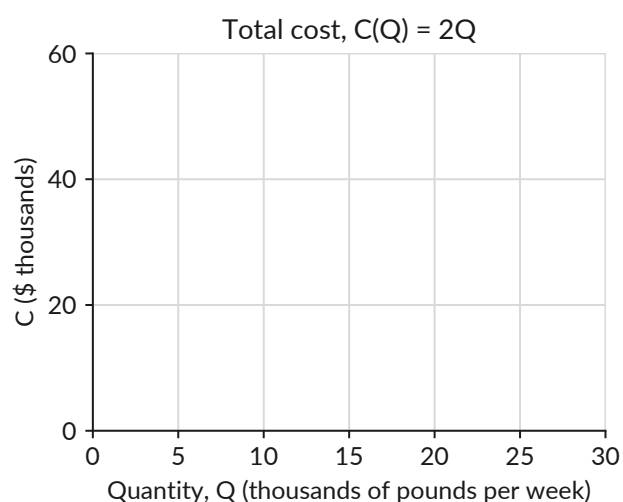
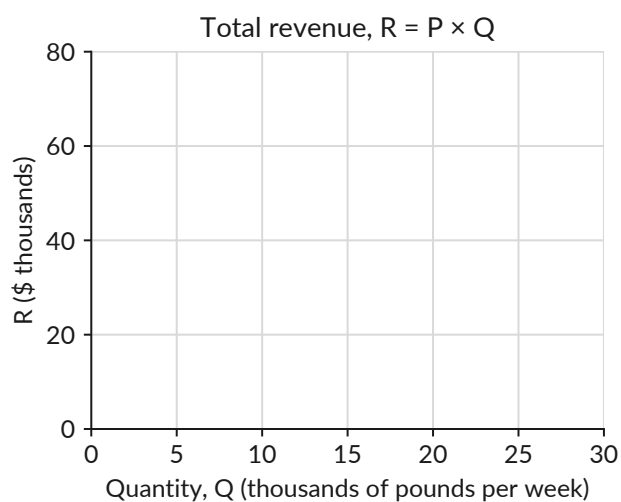
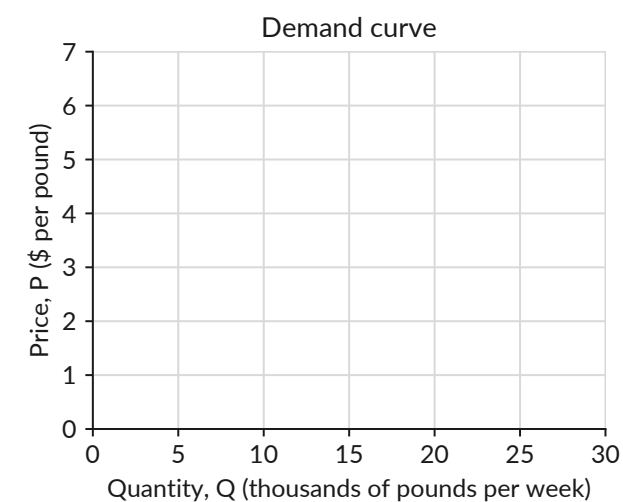
2. Pricing Apple Cinnamon Cheerios

You are the manager at General Mills who sets the price of Apple Cinnamon Cheerios. Producing the cereal costs \$2.00 per pound, so producing Q pounds costs $C(Q) = 2Q$. Market research, based on the demand curve the economist Jerry Hausman estimated for this cereal, tells you how many pounds a typical city buys each week at each price:

Price per pound, P	\$3.00	\$3.50	\$4.00	\$4.50	\$5.00
Pounds sold per week, Q	25,000	21,000	17,000	13,000	9,000
Total revenue, R					
Total cost, C					
Profit					

- (a) Fill in total revenue, total cost, and profit at each price.
- (b) Which price would you set, and what is the weekly profit there?

- (c) Plot the five columns of your table on the four grids below, one point per column on each grid, and connect the points: the demand curve (P against Q), total revenue R , total cost C , and profit.



- (d) Total revenue is highest at \$3.00, yet that is not the most profitable price. Explain why in a sentence or two.

- (e) Why does profit fall when you raise the price from \$4.00 to \$5.00, even though you earn a bigger margin on every pound?
